5.0 IN-LAB IMPEMENTATION & MEASUREMENTS (5 marks in total):

Part I: The Frequency Response Associated With [One Pole-One Zero] Transfer Function

(a) Step 1: Connect the circuit shown in **Figure 1.0.**

Connect Channel (1) of the oscilloscope to display $v_i(t)$ and Channel (2) of the oscilloscope to display $v_o(t)$.

Set the trigger source→Channel (1) and Acquire Mode→Averaging.

Adjust the controls of the function generator to provide a **sinusoidal** input voltage $v_i(t)$ of **5V (peak)** at a frequency of 100Hz.

Function Generator EDU33211A: Waveform: Sine wave, Amplitude: 5 VPP, Frequency: 100 Hz*

*Function Generator*GFG8216A: Connect to the Scope, and use Meas to check the amplitude*

(b) Step 2: Use the oscilloscope displays to measure the phase angle $\angle H(\omega)^o$ in degrees, and use either oscilloscope or DMM to measure V_i and V_o. Evaluate the magnitude $|H(\omega)|$ in dB as:

$|H(\omega)|(\text{in dB}) = [V_o(\text{in dB}) - V_i(\text{in dB})]$.

Record your results in **Table 1.0.**

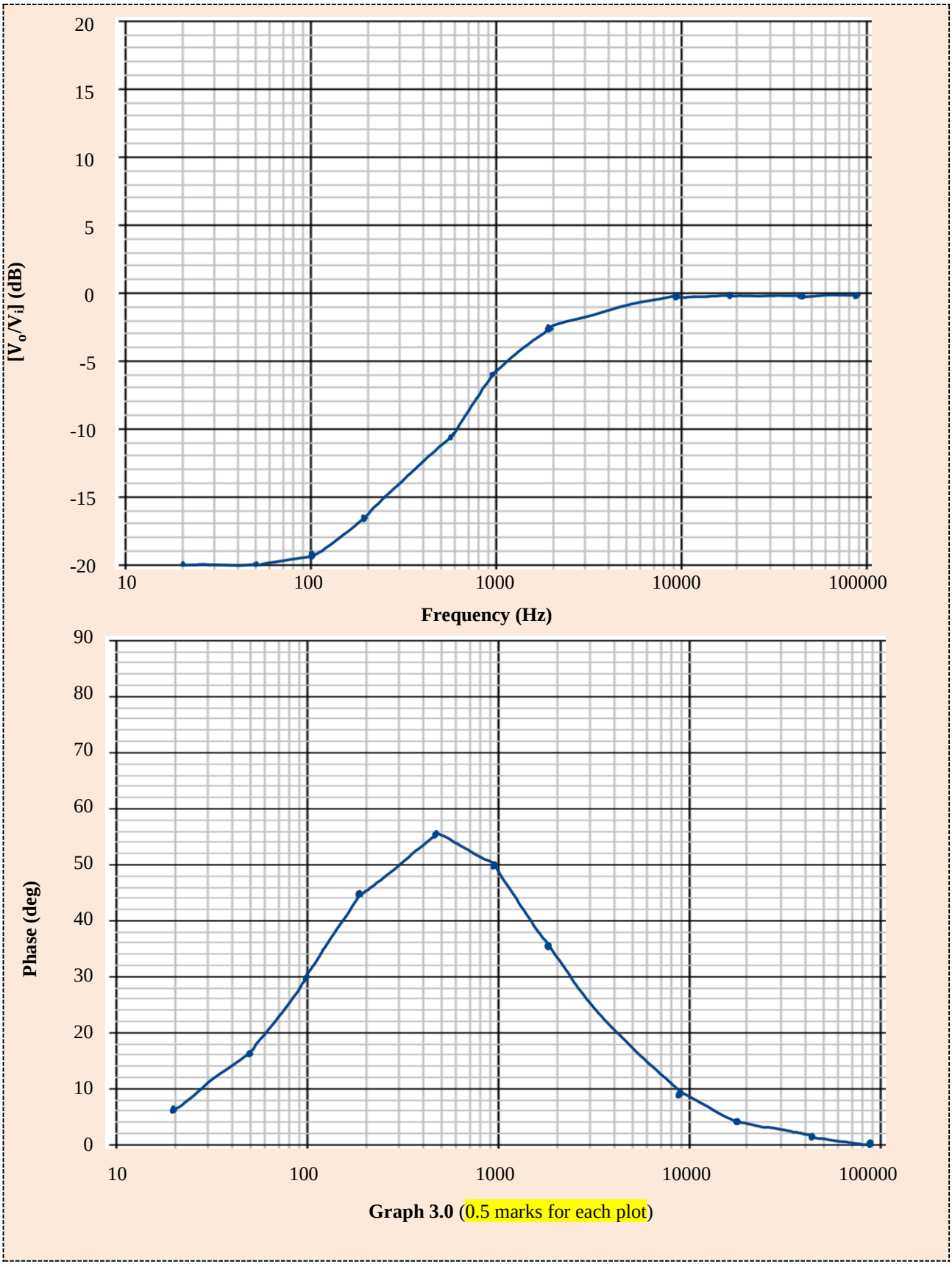
(c) Step 3: Repeat as in Step 2 for each frequency setting in **Table 1.0.**

(d) Step 4: Use **Graph 3.0** to plot the magnitude $|H(\omega)|$ in dB and phase $\angle H(\omega)$ in degrees versus frequency in Hz. Use your plot to determine the locations of the corner frequencies: f_Z and f_P. Mark these frequencies on **Graph 3.0.**

(e) Step 5: Demonstrate Step 1 to Step 4 to your TA. (1 mark)

Table 1.0 (0.5 marks)

Frequency (Hz)	$\|V_o\|$ (dB)	$\|V_i\|$ (dB)	$\|H(\omega)\|$ (dB)	$\angle H(\omega)$ (degrees)
20	-6.783	13.97	-20.75	6.4°
50	-6.375	13.97	-20.346	16.1°
100	-5.161	14.02	-19.184	29.1°
200	-2.355	14.057	-16.412	44.5°
500	3.531	14.049	-10.518	55.1°
1K	8.165	14.049	-5.884	49.3°
2K	11.573	14.049	-2.476	35.1°
10K	13.839	14.031	-0.192	9.0°
20K	13.936	14.031	-0.095	4.4°
50K	13.901	13.988	-0.087	1.4°
100K	13.927	14.005	-0.078	0.7°



Graph 3.0 (0.5 marks for each plot)

Part II: The Frequency Response Associated with [Two Poles-One Zero] Transfer Function

(f) Step 6: Connect the circuit shown in **Figure 2.0.** Connect Channel (1) of the oscilloscope to display $v_i(t)$ and Channel (2) of the oscilloscope to display $v_o(t)$; set the trigger source→Channel (1). Adjust the controls of the function generator to provide a **sinusoidal** input voltage $v_i(t)$ of **0.5V (peak)** at a frequency of 100Hz.

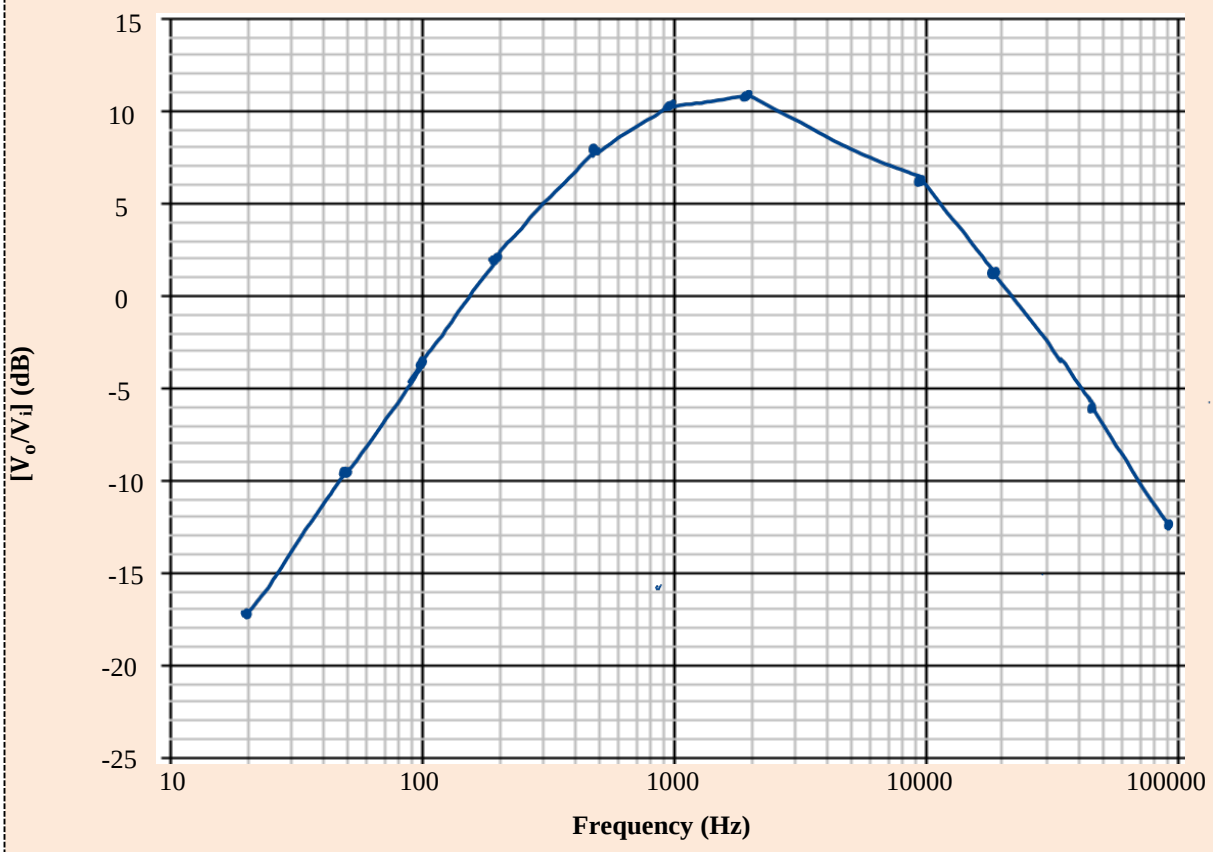
(g) Step 7: Repeat the measurements as in Step 2 and Step 3, and record your results in **Table 2.0.**

(h) Step 8: Use **Graph 4.0** to plot the magnitude $|H(\omega)|$ in dB and phase $\angle H(\omega)$ in degrees versus frequency in Hz. Use your plots to determine the locations of the corner frequencies: f_{P1} and f_{P2}. Mark these frequencies on **Graph 4.0.**

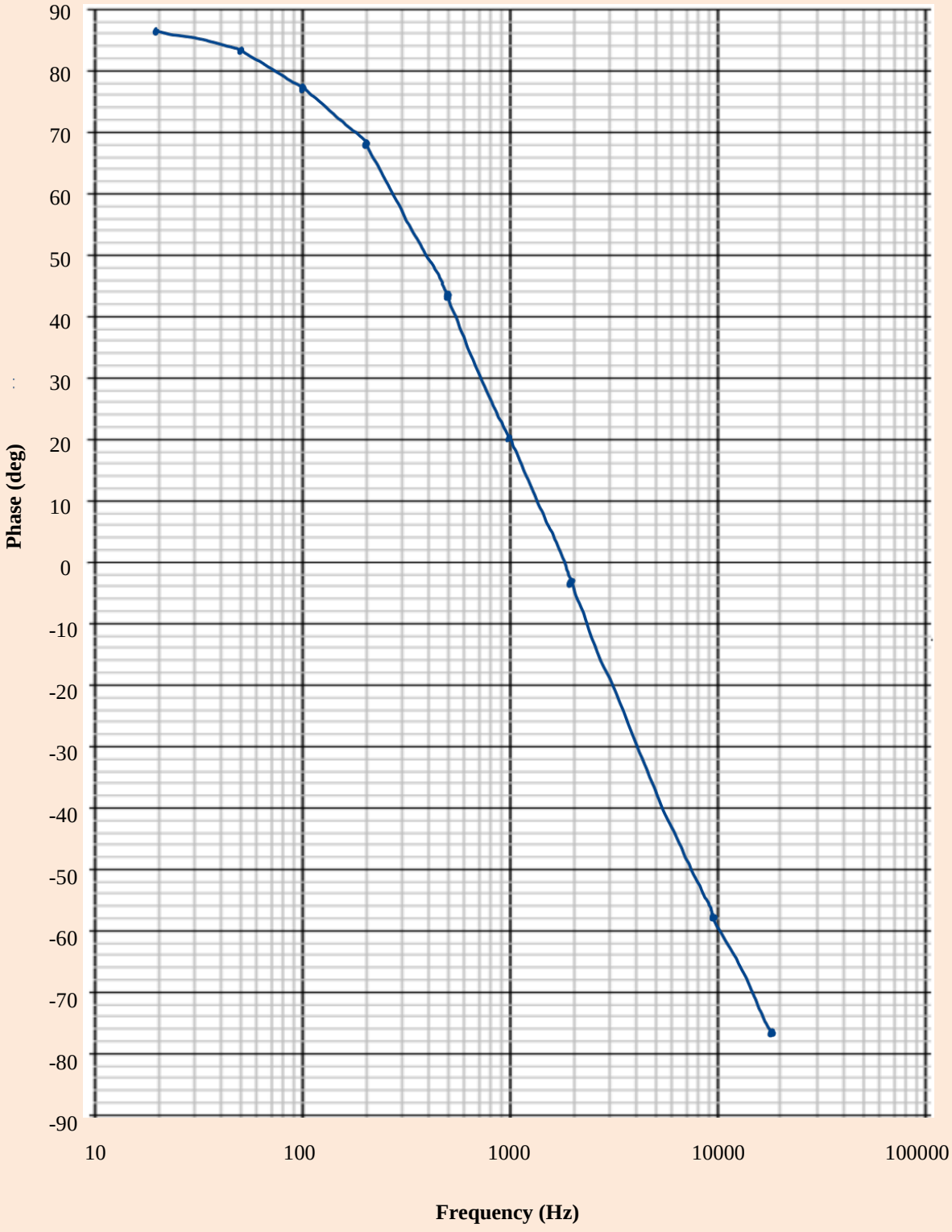
(i) Step 9: Demonstrate Step 6 to Step 8 to your TA. (1 mark)

Table 2.0 (0.5 marks)

Frequency (Hz)	$\lvert V_o \rvert$ (dB)	$\lvert V_i \rvert$ (dB)	$\lvert H(\omega) \rvert$ (dB)	$\angle H(\omega)$ (degrees)
20	-23.198	-6.012	-17.186	86.8°
50	-15.335	-5.995	-9.34	83.6°
100	-9.447	-5.969	-3.479	78.2°
200	-3.755	-5.934	2.179	68.2°
500	2.161	-5.925	8.087	44.1°
1K	4.524	-5.925	10.45	20.8°
2K	5.177	-5.925	11.103	-2.1
10K	0.407	-6.012	6.419	-56.7°
20K	-4.524	-6.055	1.531	-75.2°
50K	-12.146	-6.108	-6.038	-93.0°
100K	-18.434	-6.108	-12.326	-108.2°



Graph 4.0 *Gain* (0.5 marks)



Graph 4.0 *Phase* (0.5 marks)

6.0POST-LAB QUESTIONS (2 marks in total, 2/3 marks for each question):

(1) By considering all the asymptotic & measured plots, answer the following:
   a) At which frequencies does the asymptotic plot yield the maximum error? What would you do to minimize this error?
   b) At what frequency range does the error become negligible?

a) The plot yielded maximum errors at the corner frequencies. The magnitude was off by about 3dB right at the corner and the phase amplitude was off by about 5.7° at f_c and $10f_c$. To minimize the error, I would take extra measurements near the corners to get more accurate results.

b) The error is negligible where the frequencies are more than a decade away from the corner frequencies. For example, in Graph 3.0, the points below 16Hz and above 17.5 Hz were within 0.1 dB for magnitude and 5° for phase. The only exception is for high frequencies in the phase plot in Graph 4.0 as there is an extra pole from the op-amp's finite gain bandwidth that the ideal model did not consider.

(2) What is the maximum rate of attenuation of $|H(\omega)|$ in dB/decade for your plots on in **Graph 3.0**?

From 200 Hz to 1 kHz, $|H|$ changes from -16.4 dB to -5.9 dB which is a rate of attenuation of about 15 dB/decade.

(3) By considering your plots on **Graph 4.0,** answer the following:

a) What is the maximum rate of attenuation of | H(ω)| in dB/decade?

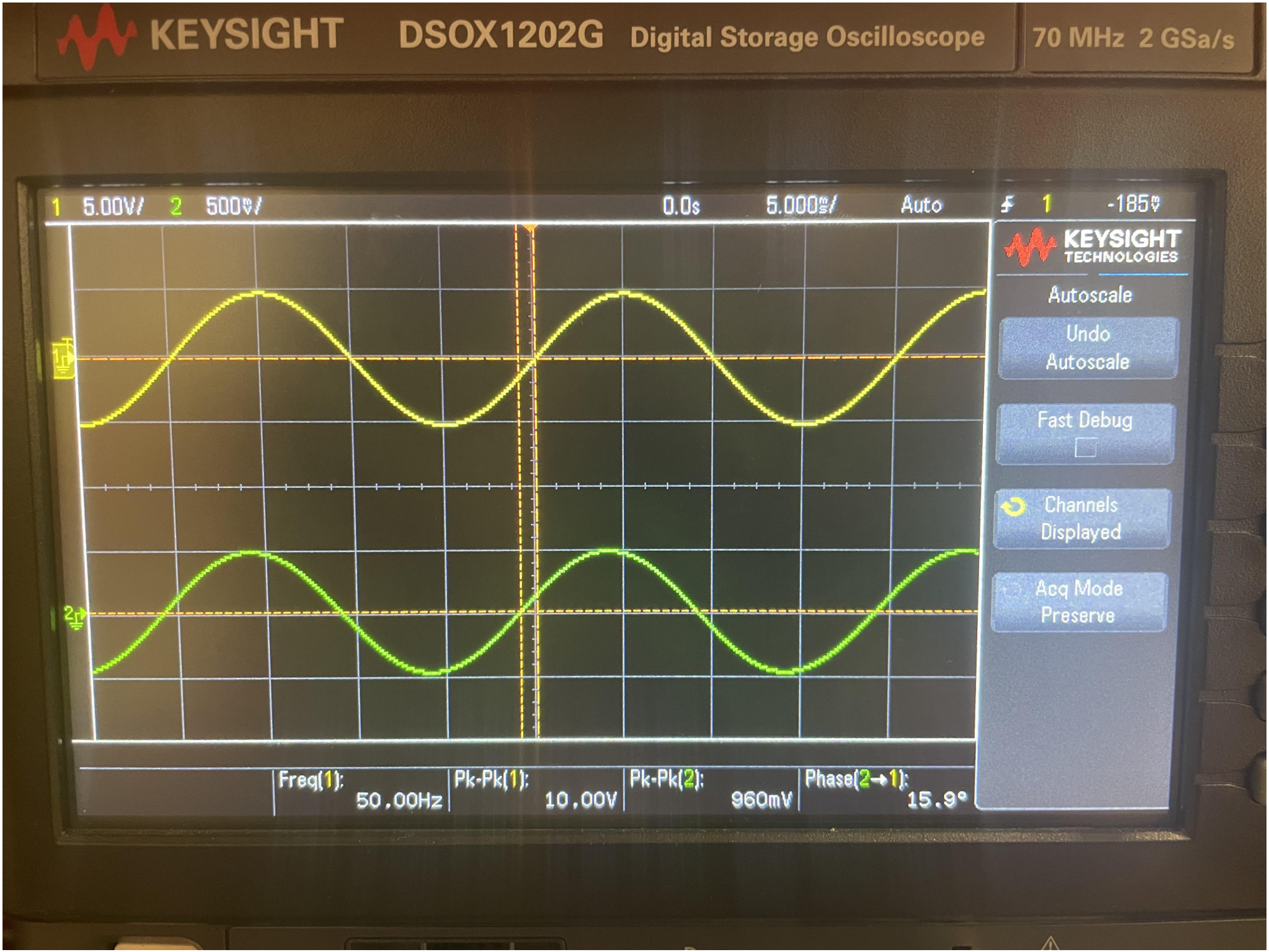
b) Suppose that the 2.7kΩ resistor and the 0.01μF capacitor at the input part of the circuit in **Figure 2.0** were interchanged, what effect would this have on the frequency-response characteristics?

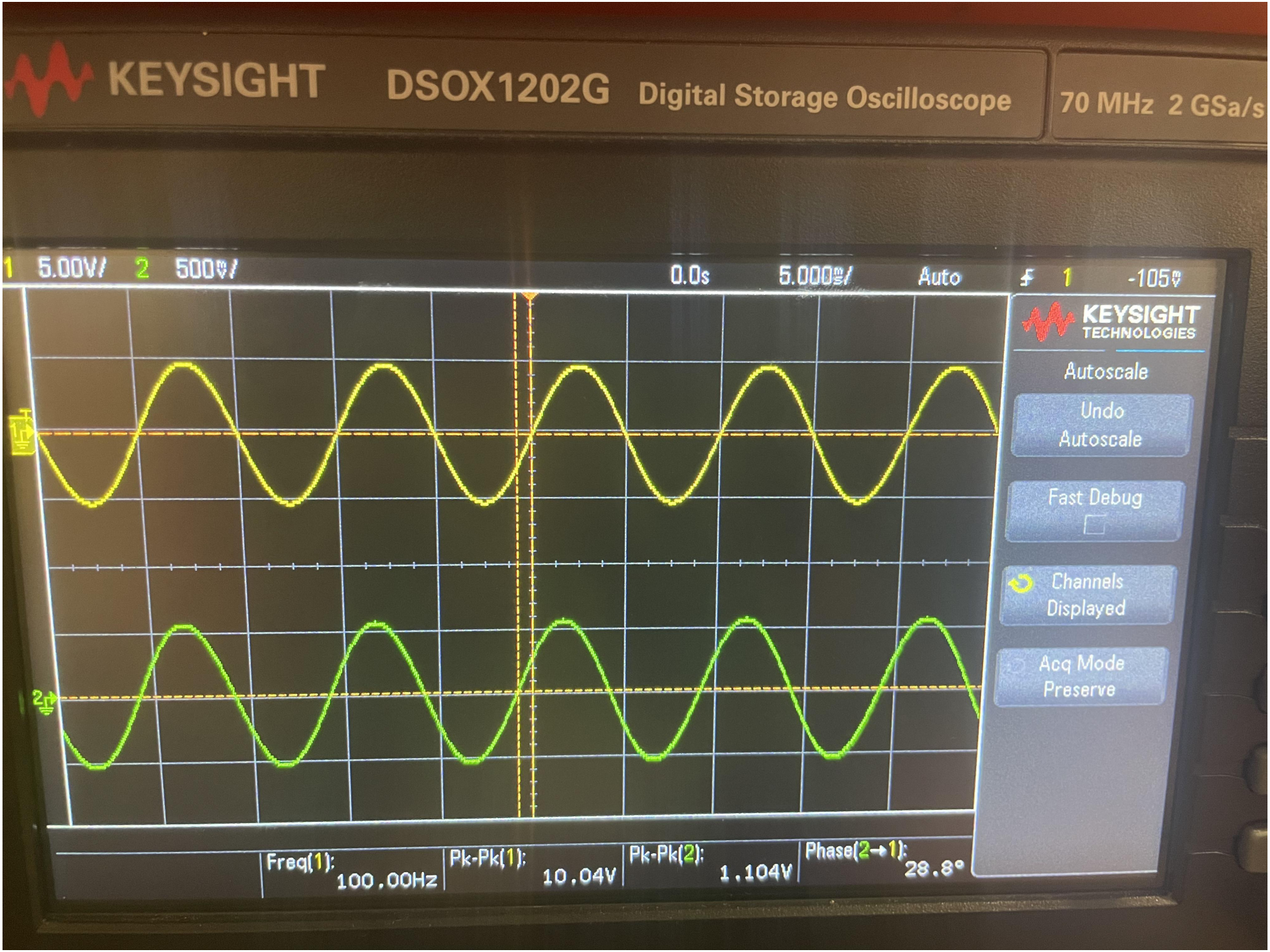
a) From 20kHz to 100kHz, |H| changes from 1.53 dB to -12.33dB which is a rate of attenuation of about 19.8 dB/decade.

b) By interchanging the 2.7kΩ resistor and 0.01μF capacitor, the input network changes from a RC low-pass to a RC high-pass where $\frac{V_o}{V_i} = \frac{sR_1C_1}{1+sR_1C_1}$ and making the new transfer function $H(s) = \frac{4.03(s/3704)(s/37037)}{(1+s/3704)(1+s/37037)}$ (2 zeros and 2 poles).

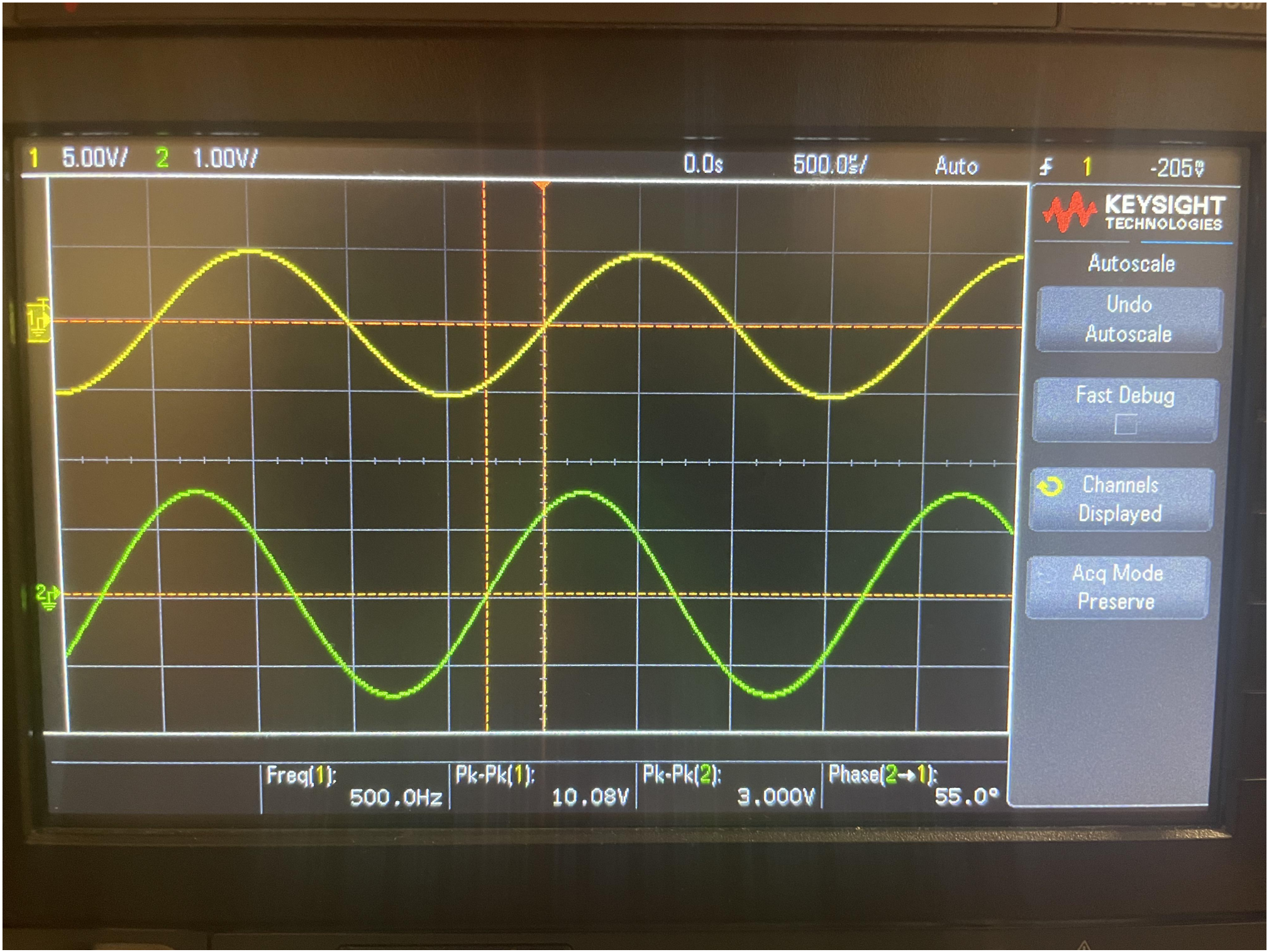
The |H| graph would have a 40 dB/decade rise under 540Hz, 20dB/decade rise between 540Hz and 5.9 kHz and a flat slope above 5.9 kHz. The phase graph would go from 180° to 0° instead of 90° to -90°. The actual corner frequencies would not change as the values of the components do not change.

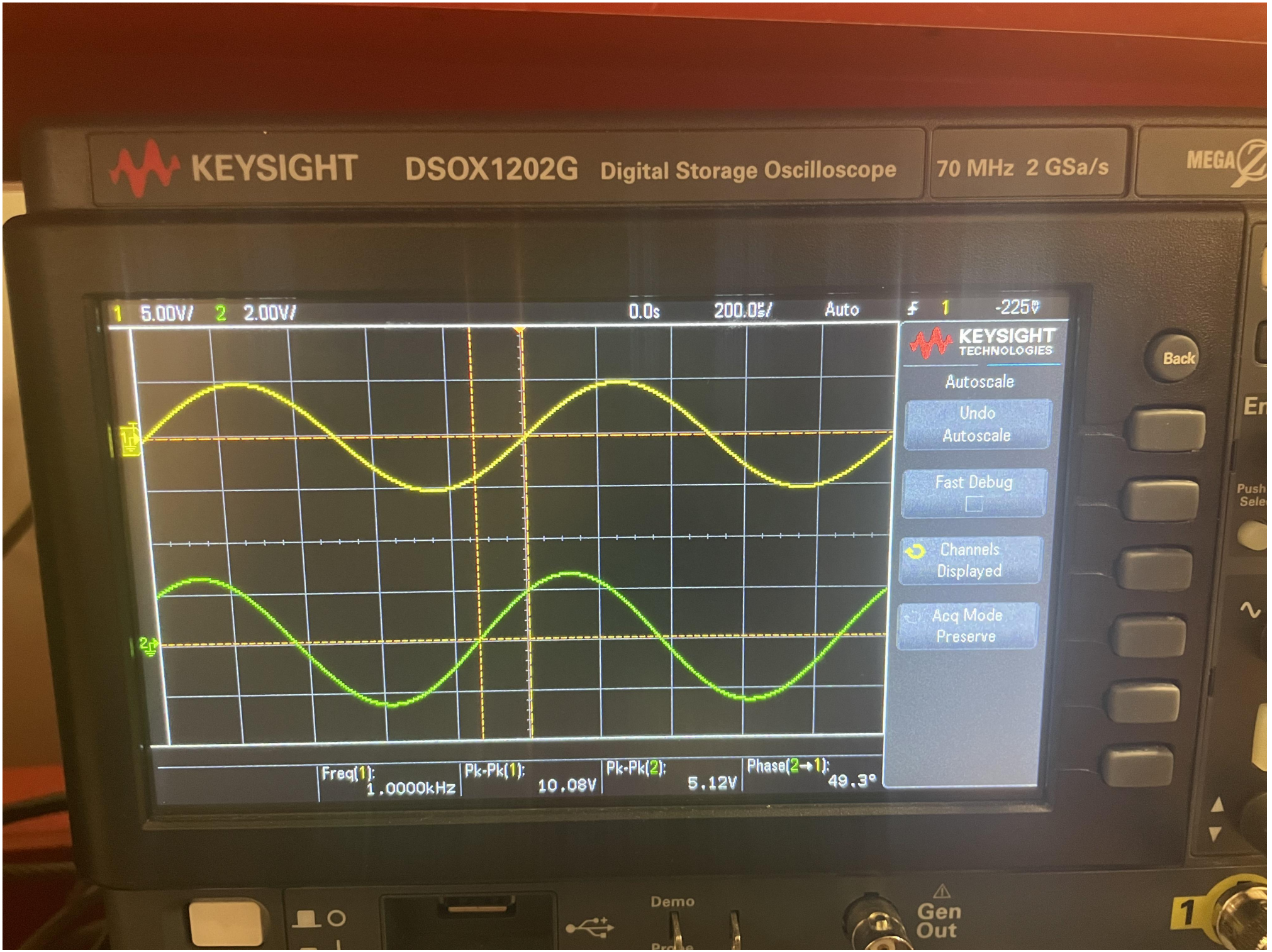
KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 GSa/s
MEGAZoom
1 5.00V/ 2 500mV/
0.0s
20.00ms/
Auto
1 -180°
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Preserve
Back
Auto Scale
Push Fine
Default Setup
Entry
Push to Select
Intensity
Math
Freq(1): 20.00Hz
Pk-Pk(1): 9.99V
Pk-Pk(2): 916mV
Phase(2→1): 6.4°
Demo
Probe Comp
Gen Out
1

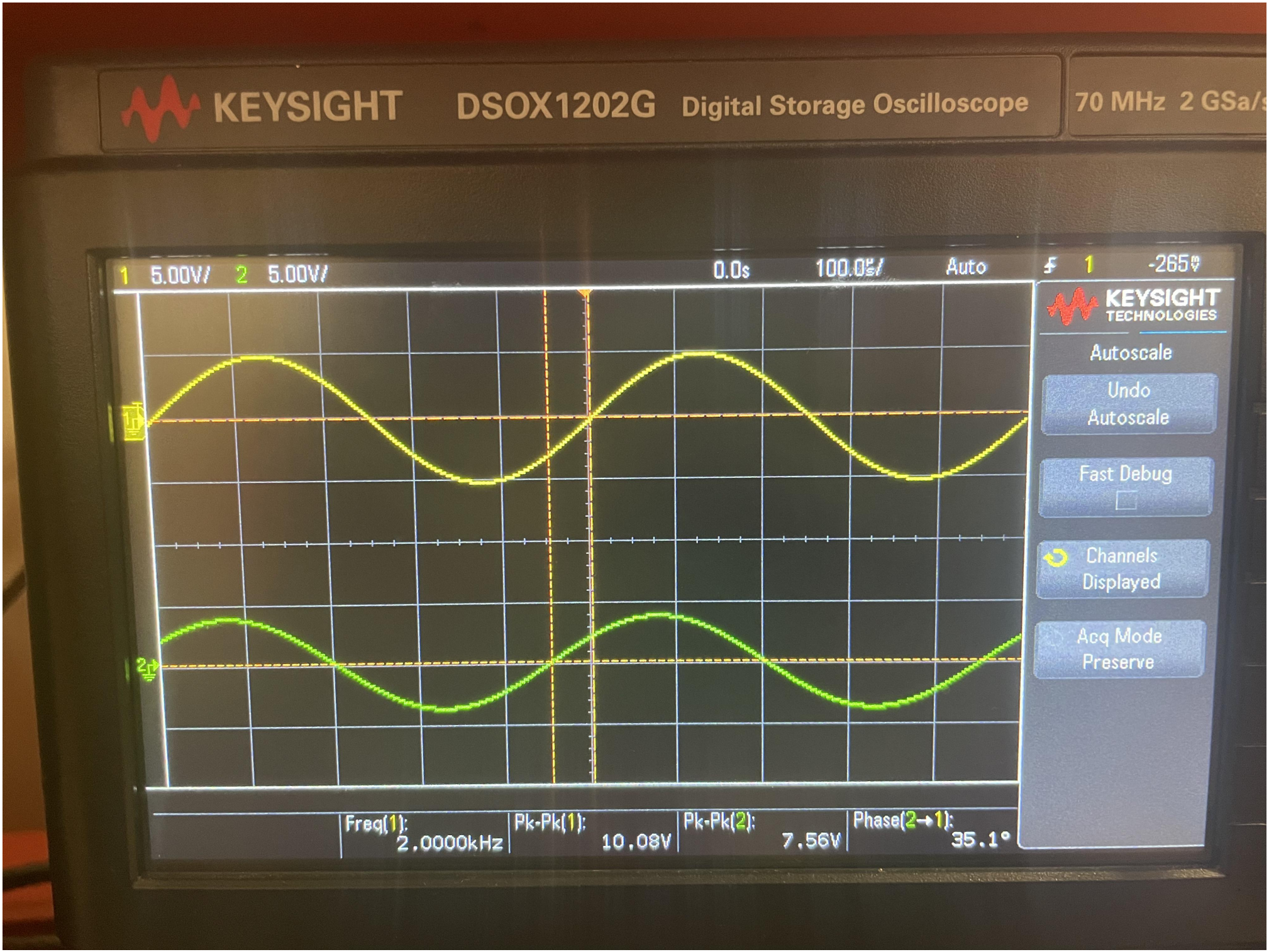

KEYSIGHT DSOX1202G Digital Storage Oscilloscope
70 MHz 2 GSa/s
1 5.00V/ 2 500mV/
0.0s
5.000ms/
Auto
1 -185mV
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Preserve
Freq(1): 50.00Hz
Pk-Pk(1): 10.00V
Pk-Pk(2): 960mV
Phase(2→1): 15.9°

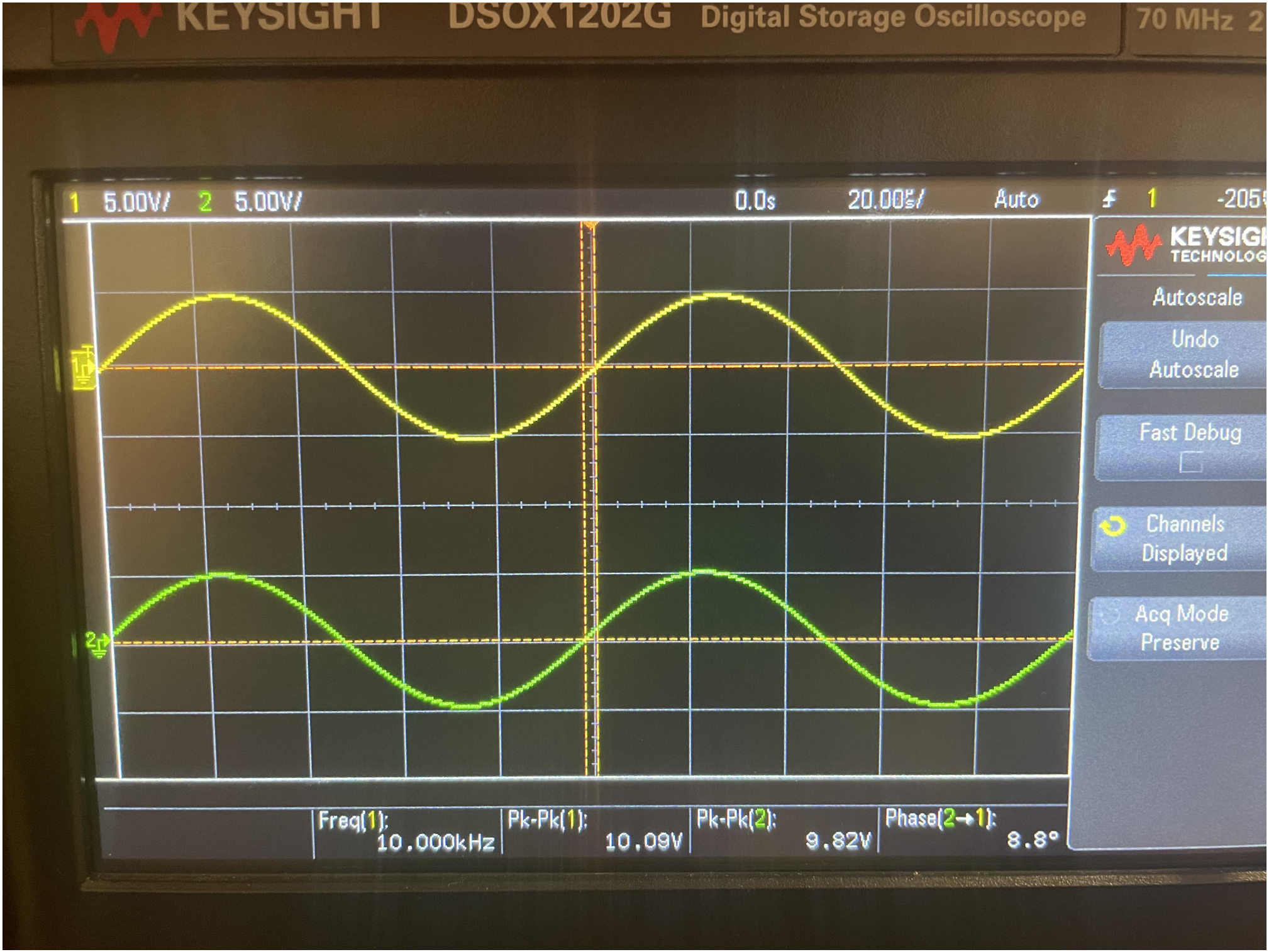

KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 GSa/s
1 5.00V/ 2 500mV/
0.0s
5.000ms/
Auto
1 -105mV
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Preserve
Freq(1): 100.00Hz
Pk-Pk(1): 10.04V
Pk-Pk(2): 1.104V
Phase(2→1): 28.8°

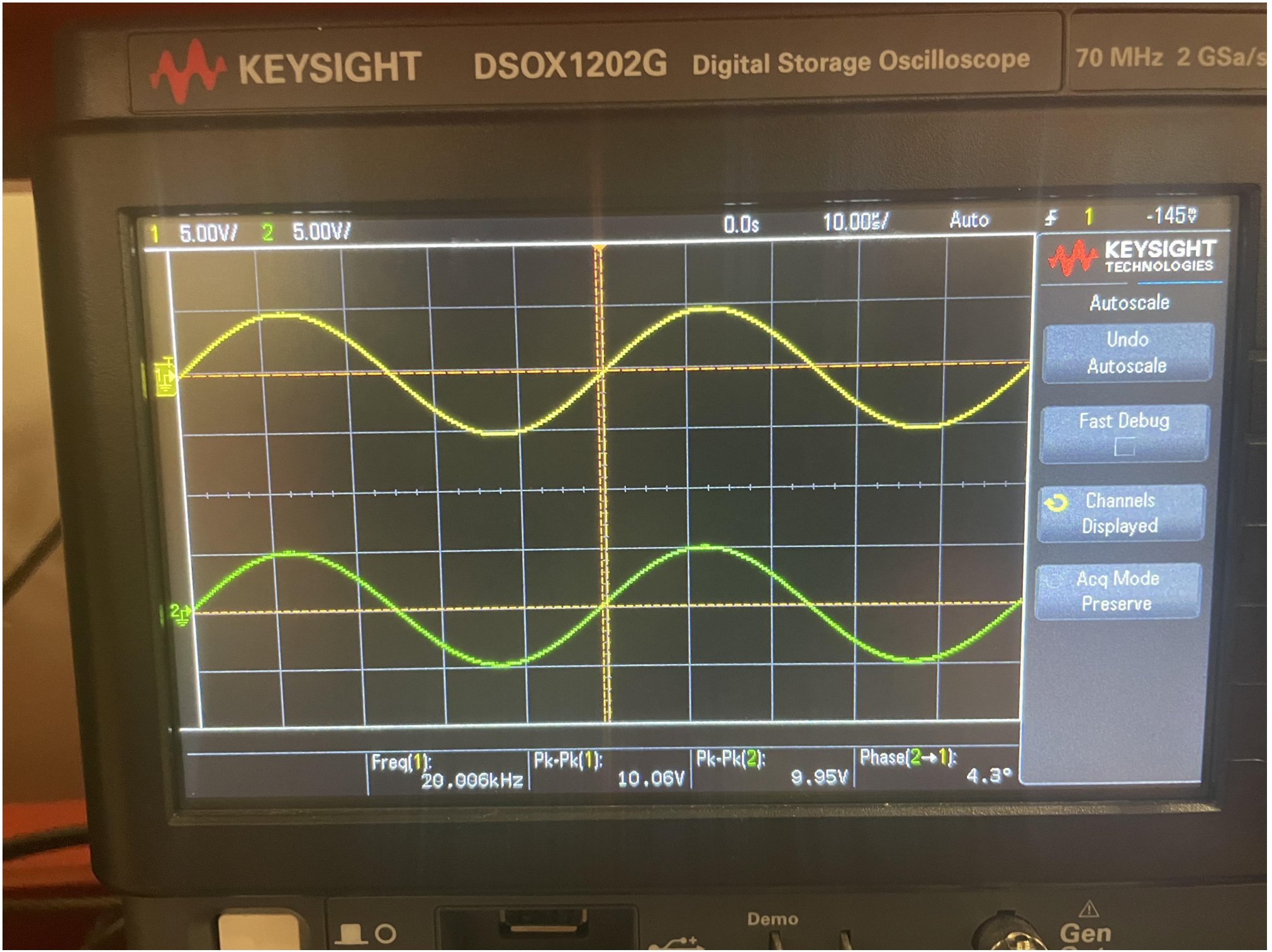
KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 GSa/s
1 5.00V/ 2 500mV/
0.0s
1.000ms/
Auto
1 -105°
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Preserve
Freq(1): 200.00Hz
Pk-Pk(1): 10.09V
Pk-Pk(2): 1.524V
Phase(2→1): 44.5°


1 5.00V/ 2 1.00V/
0.0s
500.0µs/
Auto
1 -205mV
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Preserve
Freq(1): 500.0Hz
Pk-Pk(1): 10.08V
Pk-Pk(2): 3.000V
Phase(2→1): 55.0°

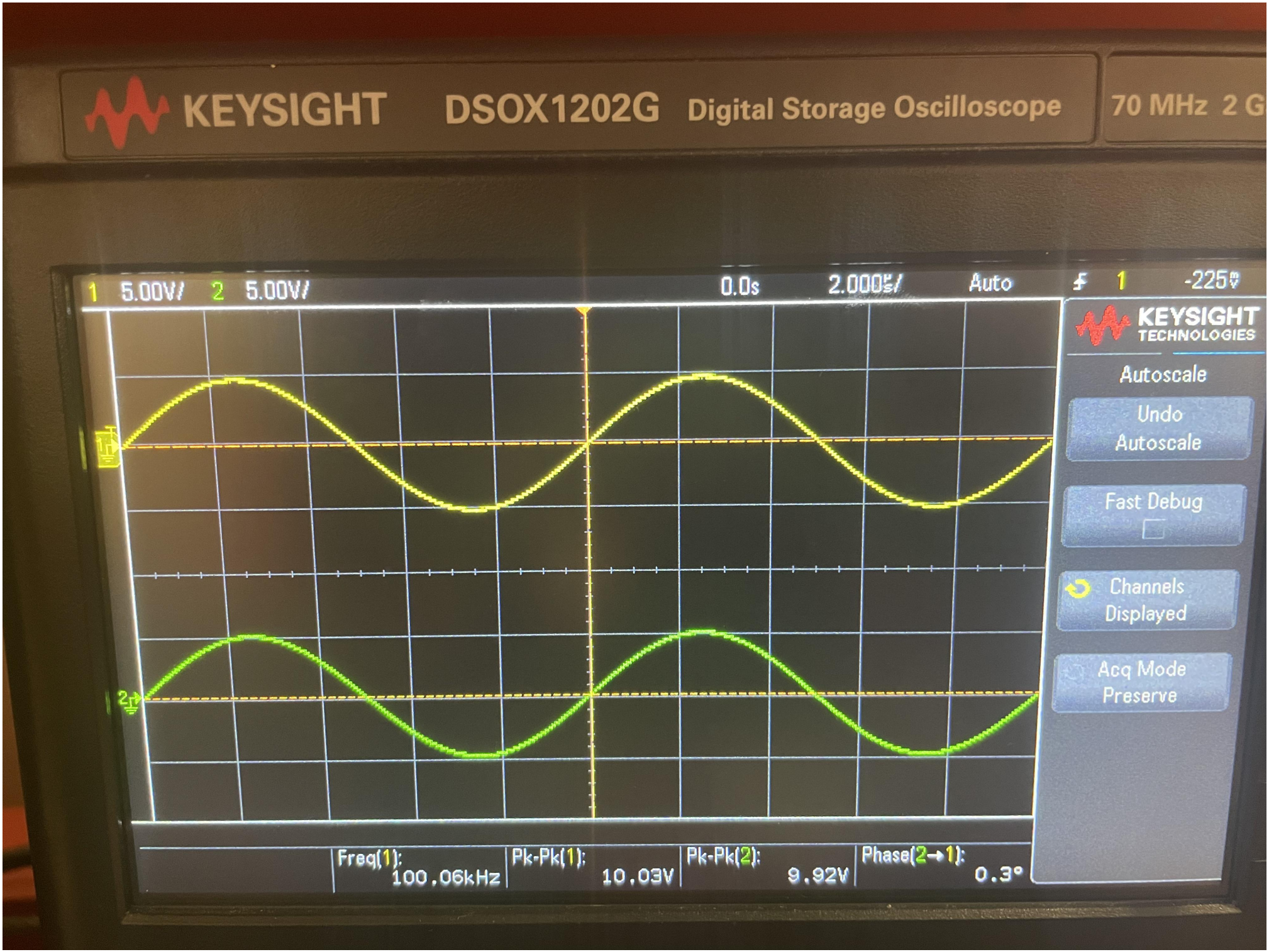

KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 GSa/s
MEGA
1 5.00V/ 2 2.00V/
0.0s
200.0ns/
Auto
1 -225°
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Preserve
Back
Freq(1): 1.0000kHz
Pk-Pk(1): 10.08V
Pk-Pk(2): 5.12V
Phase(2→1): 49.3°
Demo
Gen Out

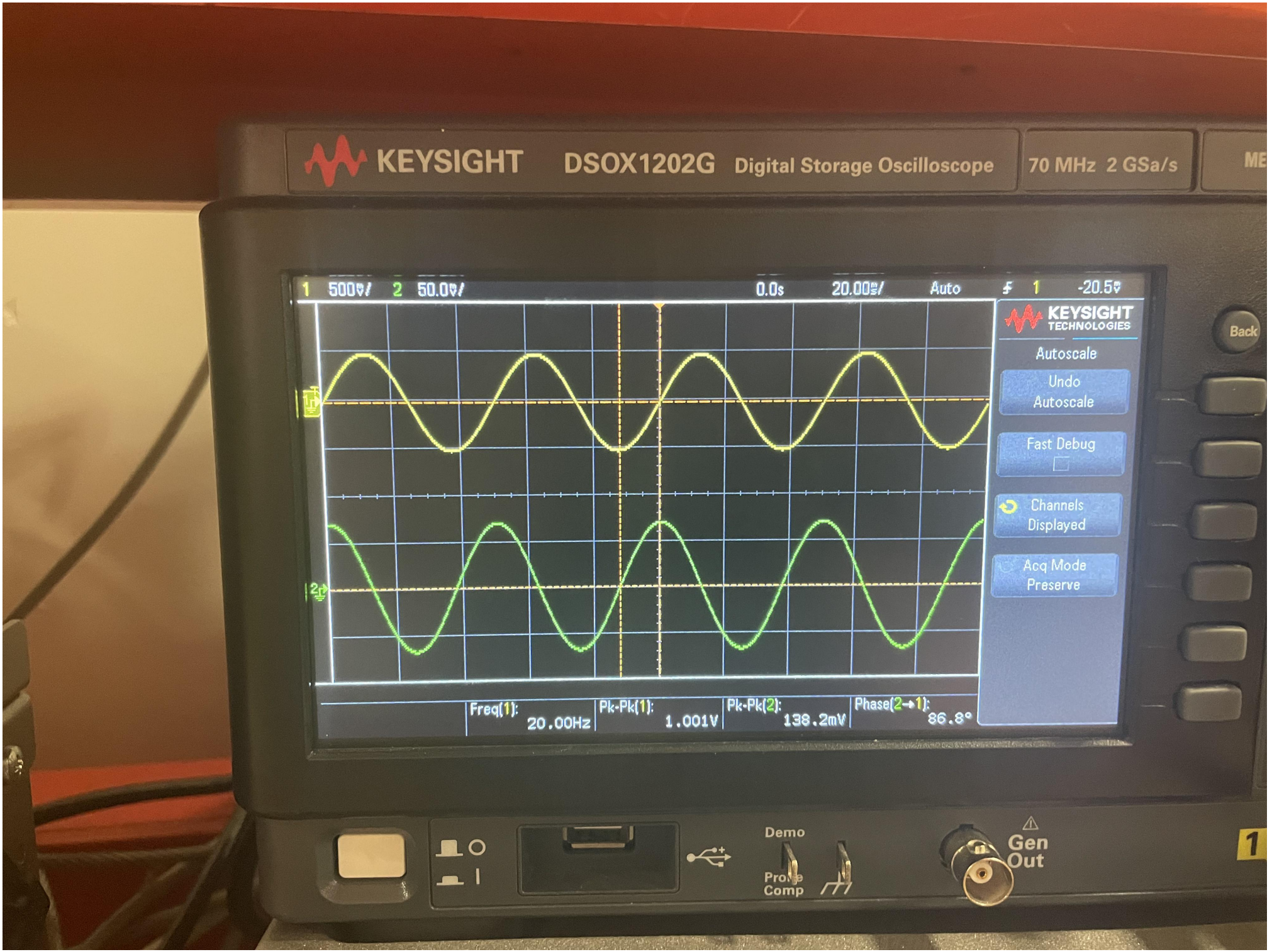

KEYSIGHT DSOX1202G Digital Storage Oscilloscope
70 MHz 2 GSa/s
1 5.00V/ 2 5.00V/
0.0s
100.0ns/
Auto
1 -265°
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Preserve
Freq(1): 2.0000kHz
Pk-Pk(1): 10.08V
Pk-Pk(2): 7.56V
Phase(2→1): 35.1°

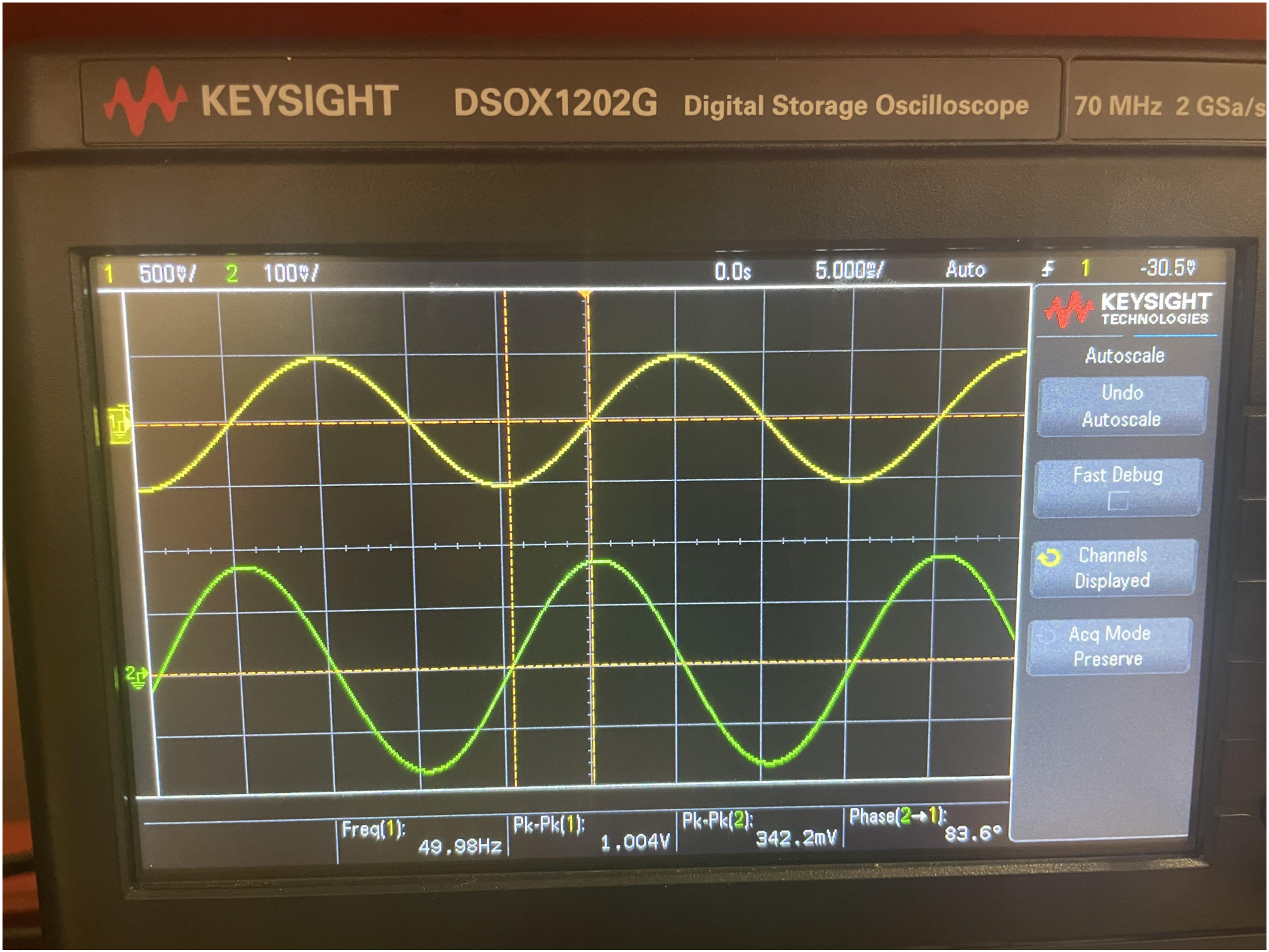

KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2
1 5.00V/ 2 5.00V/
0.0s
20.00ns/
Auto
1
-205
KEYSIGHT TECHNOLOG
Autoscale
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Preserve
Freq(1):
10.000kHz
Pk-Pk(1):
10.09V
Pk-Pk(2):
9.82V
Phase(2→1):
8.8°

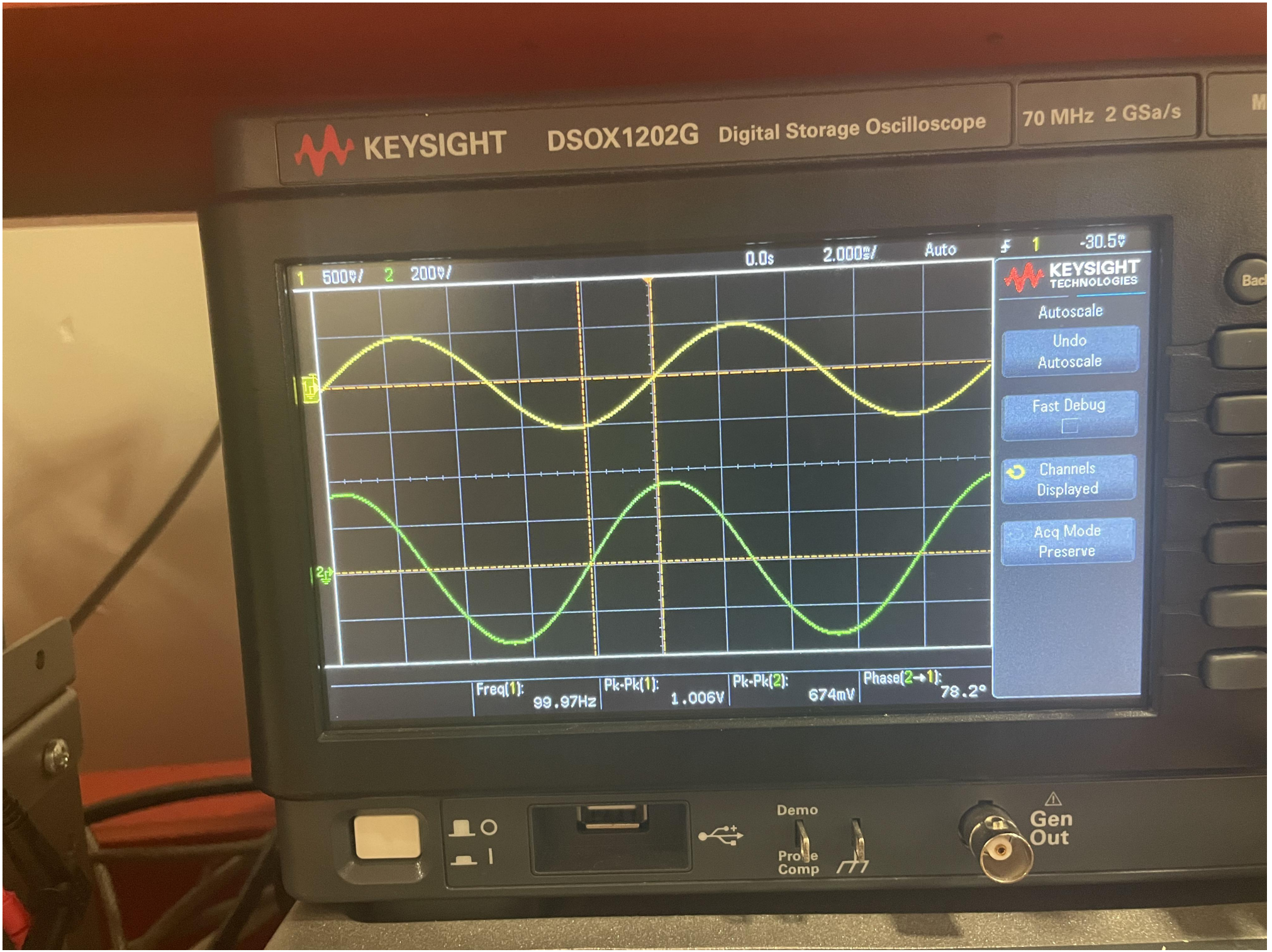

KEYSIGHT DSOX1202G Digital Storage Oscilloscope
70 MHz 2 GSa/s
1 5.00V/ 2 5.00V/
0.0s
10.00ns/
Auto
1 -145°
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Preserve
Freq(1): 20.006kHz
Pk-Pk(1): 10.06V
Pk-Pk(2): 9.95V
Phase(2→1): 4.3°
Demo
Gen

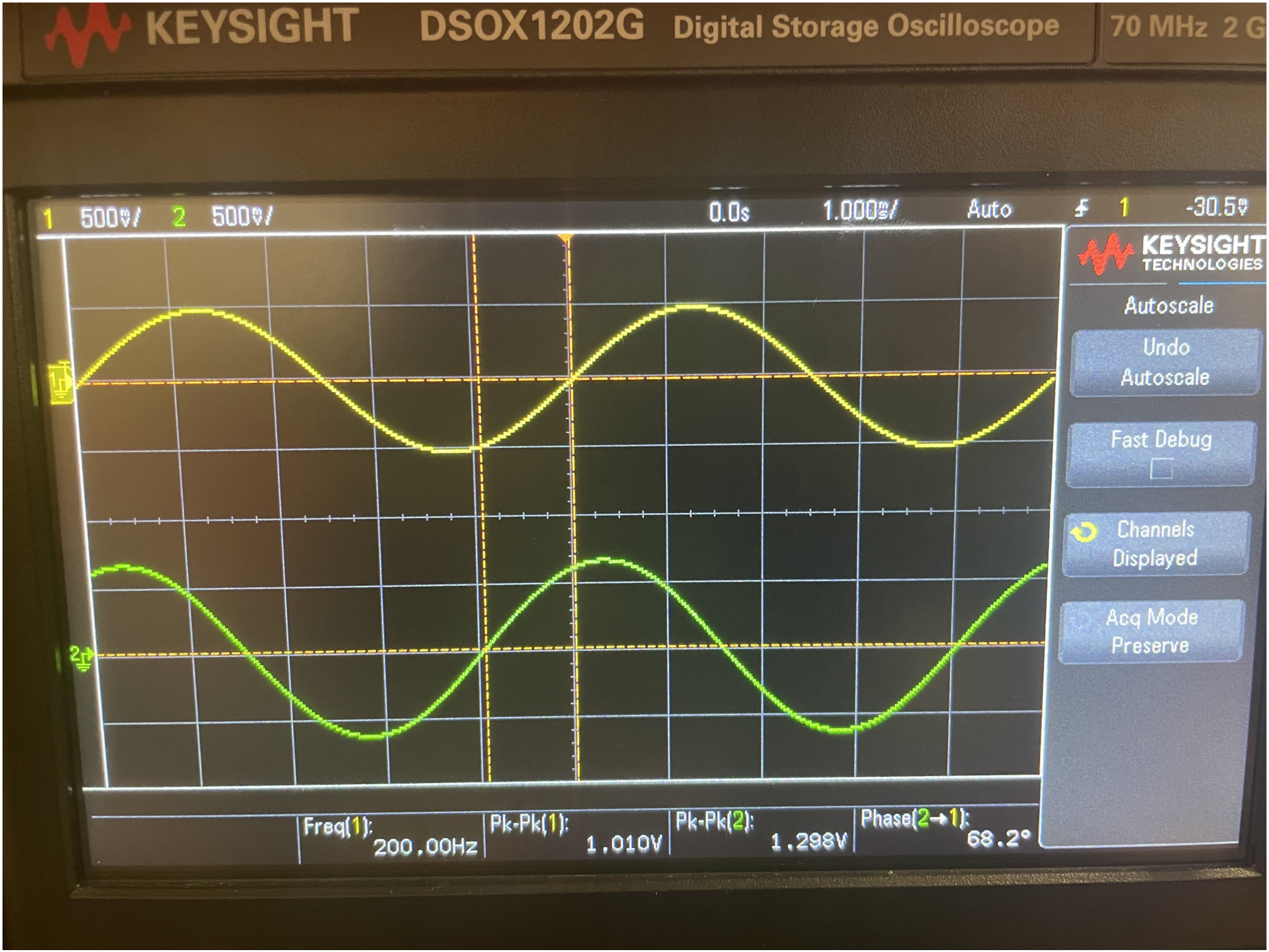
KEYSIGHT DSOX1202G Digital Storage Oscilloscope
70 MHz 2 GSa/s
5.00V/ 2 5.00V/
0.0s
5.000s/
Auto
1 -185mV
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Preserve
Freq(1): 49.99kHz
Pk-Pk(1): 10.01V
Pk-Pk(2): 9.92V
Phase(2→1): 1.2°

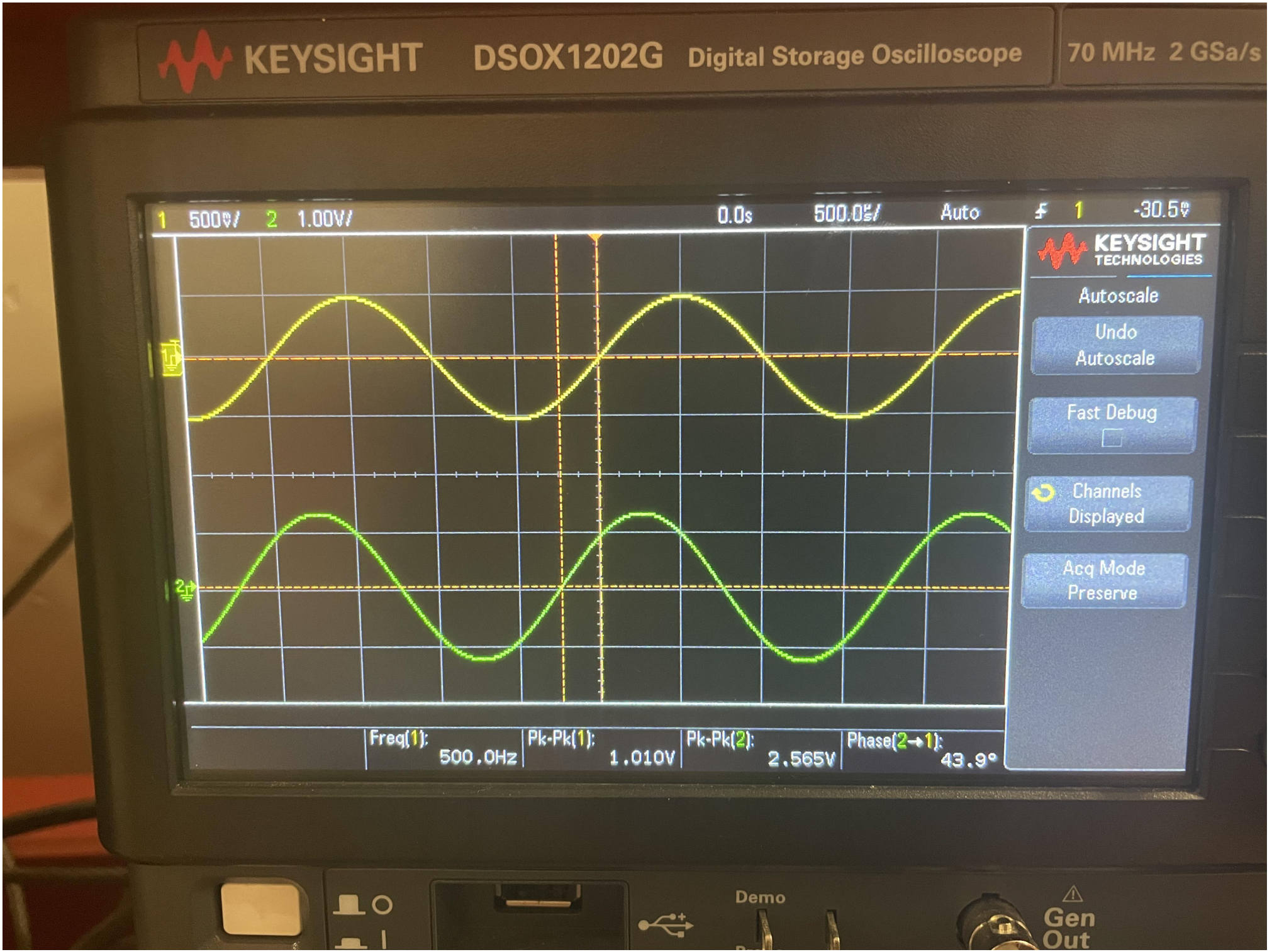

KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 G
1 5.00V/ 2 5.00V/
0.0s
2.000s/
Auto
1 -225mV
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Preserve
Freq(1): 100.06kHz
Pk-Pk(1): 10.03V
Pk-Pk(2): 9.92V
Phase(2→1): 0.3°

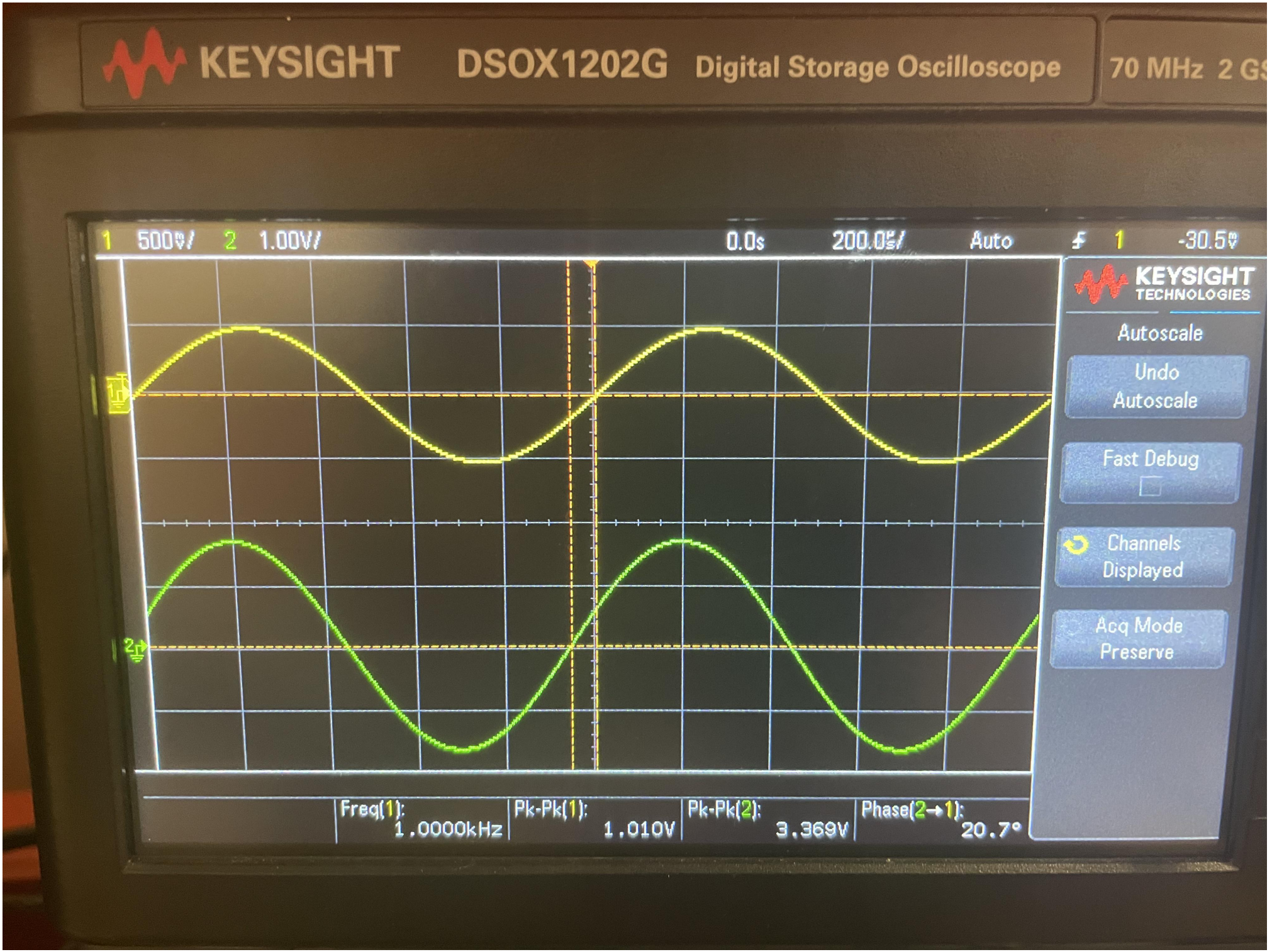

KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 GSa/s
1 500mV/ 2 50.0mV/
0.0s
20.00ms/
Auto
1 -20.5mV
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Preserve
Back
Freq(1): 20.00Hz
Pk-Pk(1): 1.001V
Pk-Pk(2): 138.2mV
Phase(2→1): 86.8°
Demo
Gen Out
1

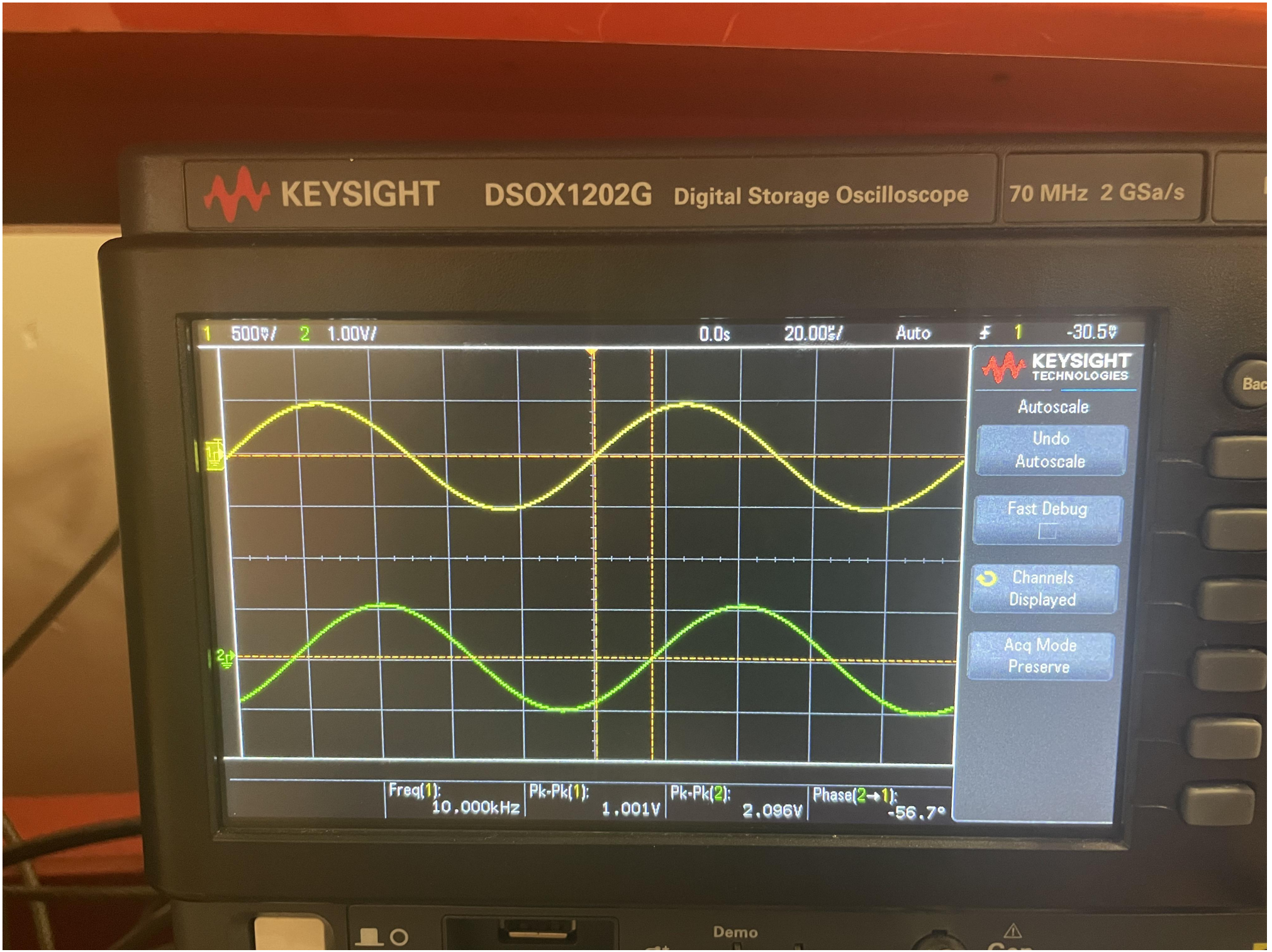

KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 GSa/s
1 500mV/ 2 100mV/
0.0s
5.000ms/
Auto
1 -30.5mV
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Preserve
Freq(1): 49.98Hz
Pk-Pk(1): 1.004V
Pk-Pk(2): 342.2mV
Phase(2→1): 83.6°

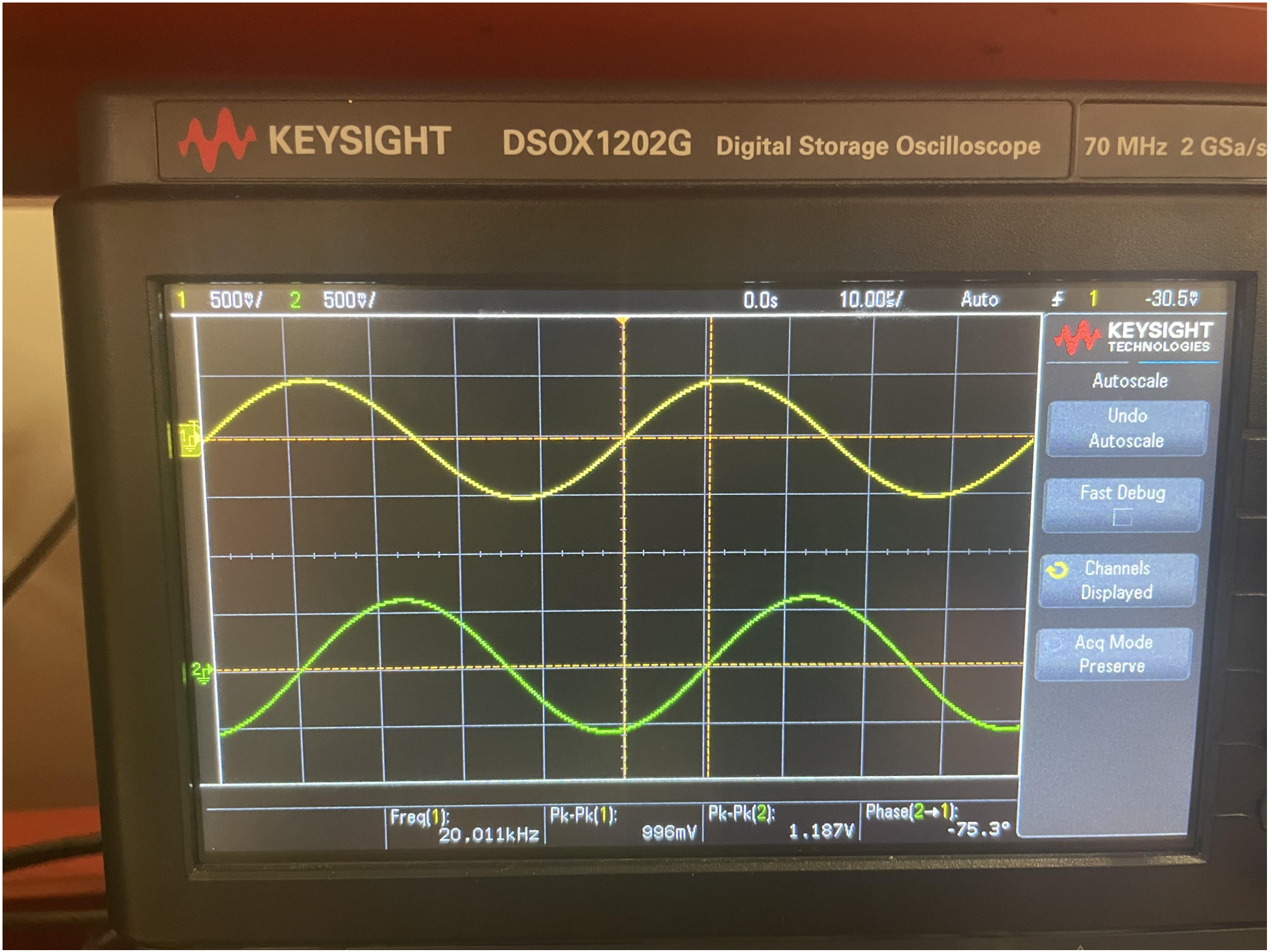

KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 GSa/s
1 500mV/ 2 200mV/
0.0s
2.000ms/
Auto
1 -30.5mV
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Preserve
Freq(1): 99.97Hz
Pk-Pk(1): 1.006V
Pk-Pk(2): 674mV
Phase(2→1): 78.2°
Demo
Probe Comp
Gen Out

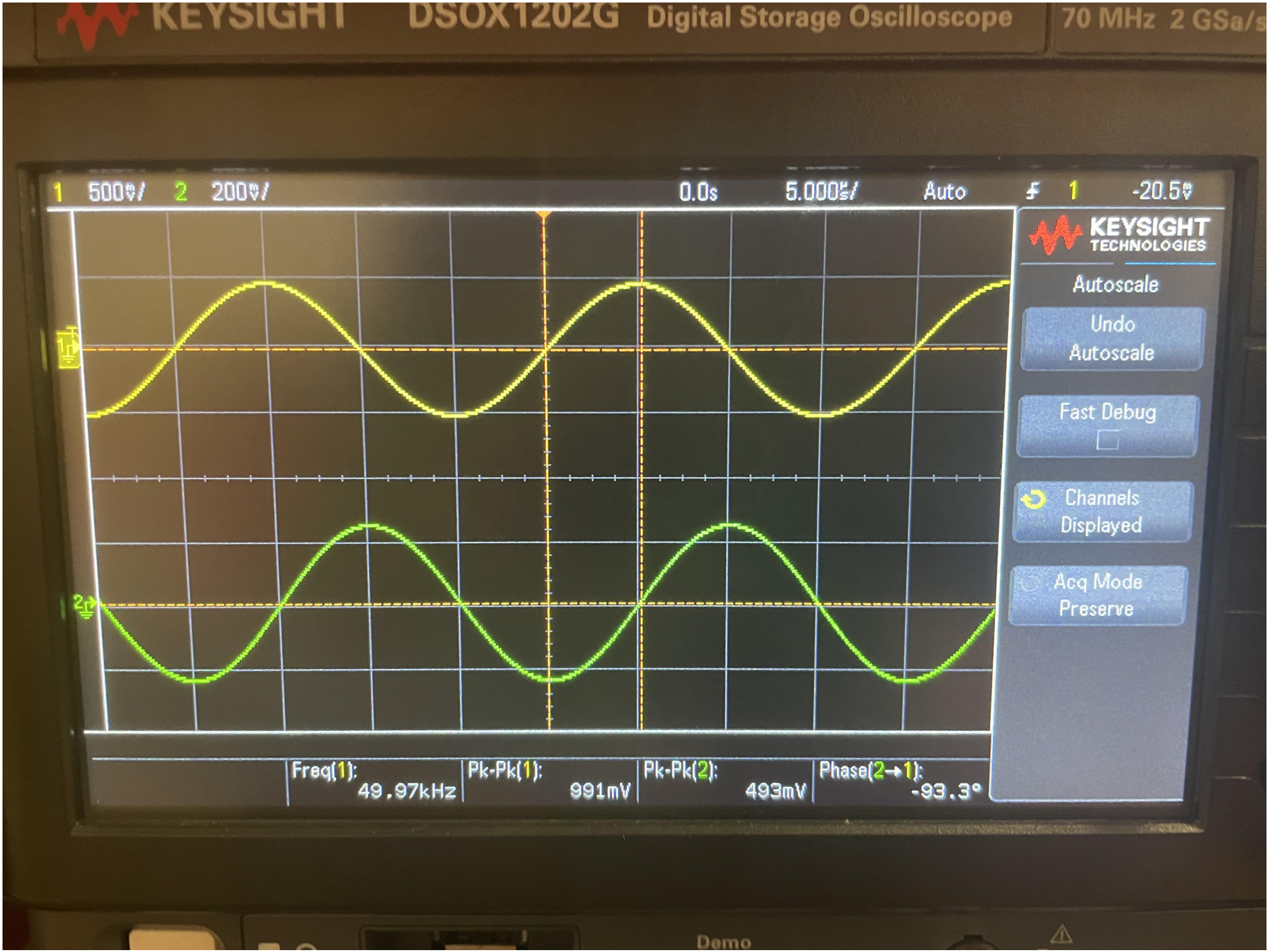

KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 G
1 500mV/ 2 500mV/
0.0s
1.000ms/
Auto
1 -30.5mV
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Preserve
Freq(1): 200.00Hz
Pk-Pk(1): 1.010V
Pk-Pk(2): 1.298V
Phase(2→1): 68.2°

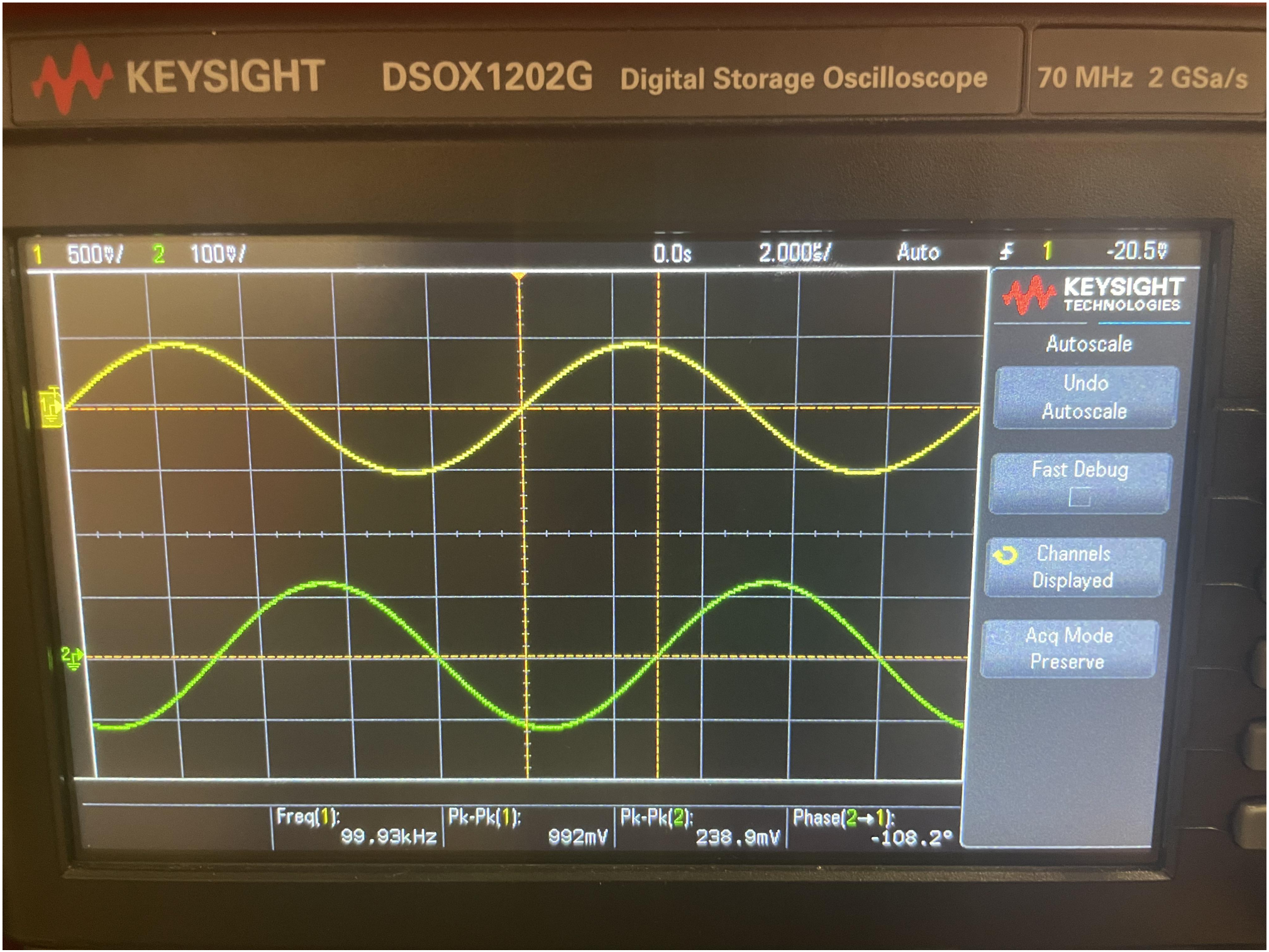

KEYSIGHT DSOX1202G Digital Storage Oscilloscope
70 MHz 2 GSa/s
1 500mV/ 2 1.00V/
0.0s
500.0µs/
Auto
1 -30.5°
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Preserve
Freq(1): 500.0Hz
Pk-Pk(1): 1.010V
Pk-Pk(2): 2.565V
Phase(2→1): 43.9°
Demo
Gen Out


KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 GS
1 500mV/ 2 1.00V/ 0.0s 200.0µs/ Auto 1 -30.5mV
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Preserve
Freq(1): 1.0000kHz
Pk-Pk(1): 1.010V
Pk-Pk(2): 3.369V
Phase(2→1): 20.7°


KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 GSa/s
1 500mV/ 2 1.00V/
0.0s
20.00ns/
Auto
1 -30.5mV
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Preserve
Freq(1): 10.000kHz
Pk-Pk(1): 1.001V
Pk-Pk(2): 2.096V
Phase(2→1): -56.7°
Demo


KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 GSa/s
1 500mV/ 2 500mV/
0.0s
10.00ms/
Auto
1 -30.5mV
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Preserve
Freq(1): 20.011kHz
Pk-Pk(1): 996mV
Pk-Pk(2): 1.187V
Phase(2→1): -75.3°


KEYSIGHT DSOX1202G Digital Storage Oscilloscope
70 MHz 2 GSa/s
1 500mV/ 2 200mV/
0.0s
5.000ns/
Auto
1 -20.5mV
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Preserve
Freq(1): 49.97kHz
Pk-Pk(1): 991mV
Pk-Pk(2): 493mV
Phase(2→1): -93.3°
Demo


KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 GSa/s
1 500mV/ 2 100mV/
0.0s
2.000µs/
Auto
1 -20.5mV
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels Displayed
Acq Mode Preserve
Freq(1): 99.93kHz
Pk-Pk(1): 992mV
Pk-Pk(2): 238.9mV
Phase(2→1): -108.2°